

Nahee Kim

M.S. STUDENT IN DATA SCIENCE · TRUSTWORTHY AI & STATE-SPACE MODELS

Incheon, Republic of Korea

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"I study internal model dynamics through interpretable signals and physics-inspired principles, with the goal of detecting behavioral shifts and enabling more reliable and controllable AI systems."

Research Profile

I study how internal model signals reflect changes in neural behavior, focusing on interpretability, reliability, and robustness. My research develops physically and geometrically grounded methods to identify failure-prone inputs under noise and distribution shifts. I aim to extend these white-box methods to foundation models and AI agents for behavioral monitoring, safety evaluation, and control.

Education

Inha University

M.S. IN DATA SCIENCE

Incheon, Republic of Korea

Sep. 2025 – Feb. 2027 (Expected)

- Expected to complete the M.S. degree in three semesters through early graduation
- Research focus: trustworthy AI, state-space models, model interpretability, and scientific imaging
- Advisor: Prof. Ji Hyun Nam
- GPA: 4.13 / 4.5

Inha University

B.S. IN ELECTRONIC ENGINEERING

Incheon, Republic of Korea

Mar. 2019 – Aug. 2025

- GPA: 3.80 / 4.5

Selected Research & Publications

RiemannMamba: Geometry-Aware Internal Time in Selective State Space Models

Submitted to NeurIPS 2026

SOLE FIRST AUTHOR

2025 – Present

- **Architectural reinterpretation:** Reinterpreted Mamba's input-dependent step size Δ not merely as a selective gate, but as a state-dependent geometry on the hidden-state space.
- **Reliability:** Accumulated geometry-weighted hidden-state changes into an internal time variable τ , aligned it with sample difficulty and error propensity, and applied it to selective prediction under distribution shifts and corruptions.
- **Interpretability:** Developed gradient-free τ -CAM to spatially visualize the internal time signal and identify image regions that induce stronger hidden-state evolution.

TST-Mamba: Plug-and-Play Selective State-Space Denoising for Non-Line-of-Sight Imaging

Submitted to Neurocomputing

SOLE FIRST AUTHOR

2025 – Present

- **NLOS noise research:** Characterized the dominant noise patterns in real NLOS measurements and developed a real-measurement-guided SPAD noise simulator tailored to transient denoising.
- **Architecture:** Designed and formalized TST-Mamba as a Mamba-based skip-connected denoising autoencoder for NLOS transients, combining an undercomplete temporal bottleneck, single-pass spatial processing in latent space, and skip-latent fusion to preserve reconstruction-critical temporal details.
- **Cross-pipeline robustness and generalization:** Demonstrated that TST-Mamba operates as a reconstruction-agnostic plug-and-play denoiser, improving downstream reconstruction without modifying or retraining each pipeline and maintaining effectiveness across simulated, unseen-category, and real NLOS measurements.

Driver Handover Readiness Estimation via Bio-Vehicle Signal Fusion and Safety-Oriented Re-decision

IEIE Summer Annual Conference

2026

LEAD RESEARCHER · PEER-REVIEWED CONFERENCE PAPER

Jun. 2026

- **Project leadership and mentorship:** Conceived the project and led it end-to-end, while supervising an undergraduate researcher through problem formulation, framework design, experimental evaluation, result analysis, and manuscript preparation.
- **Methodology:** Developed a bio-vehicle signal fusion framework combining ECG-derived driver stress estimates with vehicle-state information, and designed a reward-based re-decision policy to reduce unsafe handover approvals.
- **Validation and presentation:** Evaluated the framework through driver-held-out experiments on WESAD and OpenDriver data; the paper was selected for the conference's best-paper session and presented as a poster in Jeju, Republic of Korea.

Air Purifier Utilizing Tesla Compressor and Cyclone Dust Collector

Technology and Education · KSME

KCI JOURNAL PAPER

2024

- **Research contribution:** Implemented motor control for the Tesla compressor and independently conducted the experimental analysis and manuscript writing.

Additional Research Experience

SocialWise: Augmented-Reality-Based Social Communication Support for Children with Autism Spectrum Disorder

Inha University

UNDERGRADUATE RESEARCH ASSISTANT2024

- **Human-centered assistive AI:** Contributed to the development of an augmented-reality-based system designed to interpret social interaction cues and support social communication for children with autism spectrum disorder.
- **Speech data collection:** Conducted speech-utterance data collection with participants with autism spectrum disorder for the development of the assistive communication system.

Selected Projects

Cross-Lingual Voice-Conversion Augmentation for Low-Resource Neural Vocoding

Inha University

CAPSTONE DESIGN TEAM PROJECTDec. 2024

- **Low-resource data augmentation:** Designed three data-scarcity settings retaining 50%, 25%, and 12.5% of the original LJSpeech data, and augmented the missing portions through FreeVC-based voice conversion using speaker references from one, three, and seven language back-grounds.
- **Model training and evaluation:** Trained a full-data baseline and three augmented-data MelGAN vocoders for 3,000 epochs, and compared their convergence and speech reconstruction quality using L1 loss, waveforms, and mel-spectrograms; the 50% and 25% settings approached the full-data baseline while converging faster during early training.

Teaching Experience

Object-Oriented Programming

Inha University

TEACHING ASSISTANTSpring 2026

- Assisted students with object-oriented programming concepts, programming exercises, and the implementation and debugging of course as-signments.

Image Data Analysis

Inha University

TEACHING ASSISTANTSpring 2026

- Supported coursework on image-data processing and analysis, helping students understand computational methods, experimental workflows, and the interpretation of image-based results.

Physics Laboratory

Inha University

TEACHING ASSISTANTFall 2025

- Led full undergraduate physics laboratory sessions, including explaining experimental principles and procedures, supervising measurements and equipment use, guiding data analysis, and evaluating students’ experimental work.

High School Physics

Mirae Injae Academy

PHYSICS INSTRUCTORMay 2023 – Apr. 2024

- Taught high school students fundamental physics concepts and problem-solving methods, explaining mathematical and physical principles in an accessible and structured manner.

Honors & Awards

Oct. 2023	Bronze Award , 13th National Student Design Competition (Tesla-Compressor Cyclone Collector)	KSME
Oct. 2023	Excellence Award , 15th Creative Design Competition (Multi-Phase Hybrid Generator)	I-DREAM
Sep. 2020	Bronze Award , 2020 Capstone Design Competition (Automatic Nozzle Changer for 3D Printers)	Inha University
Jul. 2020	Grand Prize , 2020 Engineering Design Club Competition (Automatic Nozzle Changer for 3D Printers)	Inha Univ. ICEE

Skills & Research Expertise

Programming	Python, C, C++
Deep Learning	PyTorch, selective state-space models, Mamba architectures, CNNs, and spatio-temporal modeling
Trustworthy AI	Interpretability, selective prediction, reliability estimation, and robustness under corruptions
Scientific Imaging	NLOS imaging, transient-data processing, SPAD noise simulation, reconstruction, and denoising
Languages	Korean (native), English (working proficiency)

References

Prof. Ji Hyun Nam	Assistant Professor and Department Chair, Department of Statistics and Data Science, Inha University namjihyun@inha.ac.kr M.S. Advisor
Prof. Kyounghee Baek	Professor, School of Law, Inha University khbaek@inha.ac.kr Academic Referee (AI Ethics & Law Coursework)